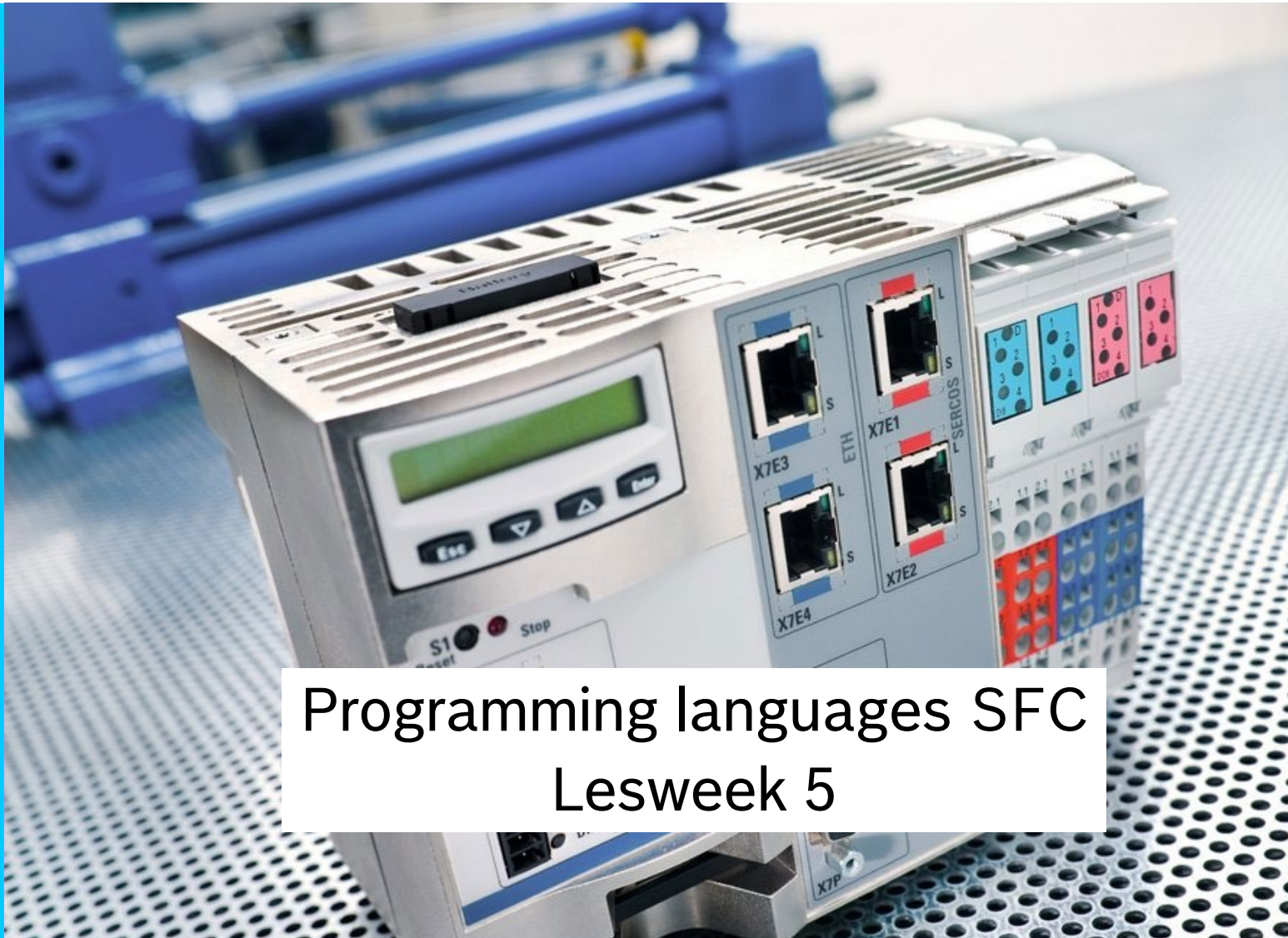




Programming languages SFC

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rexroth
A Bosch Company

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- Programming with SFC

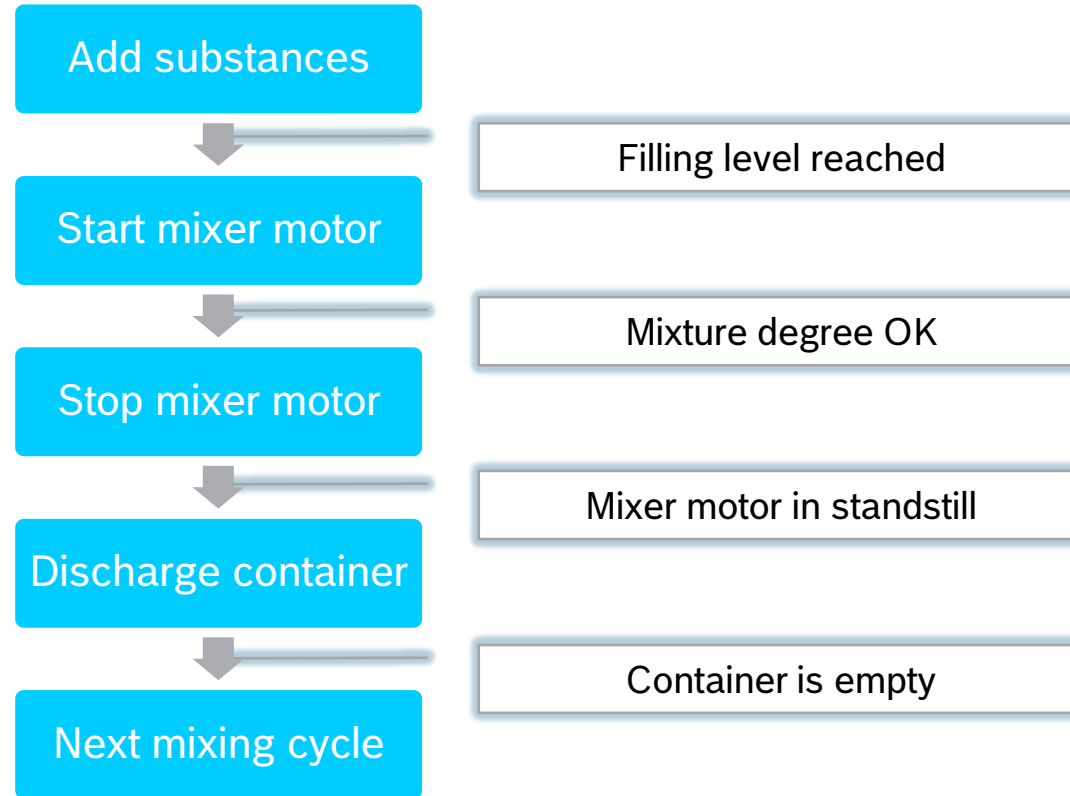
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SFC – Sequential Function Chart

- SFC has been specified to subdivide complex process into manageable units and to describe the flow of control between this units
- SFC is suited for sequential processing as well as parallel processing
- Main elements of SFC are steps and transitions
- A **step** is associated by a set of instructions, which are executed as long as the step is active
- The programming of the steps can be done in any IEC language
- It is even possible to implement a step itself again in SFC
- A **Transition** determines (Boolean expression), when the step is deactivated
- SFC is specially suited for processes with a step-by-step state machine, e.g. a chemical mixing process or a washing machine

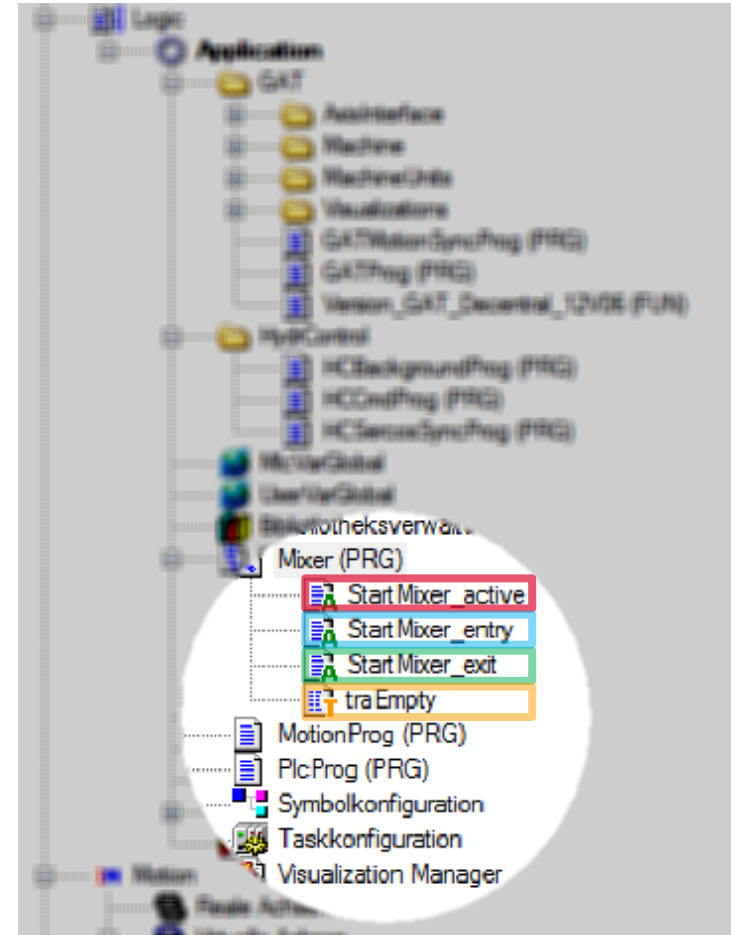
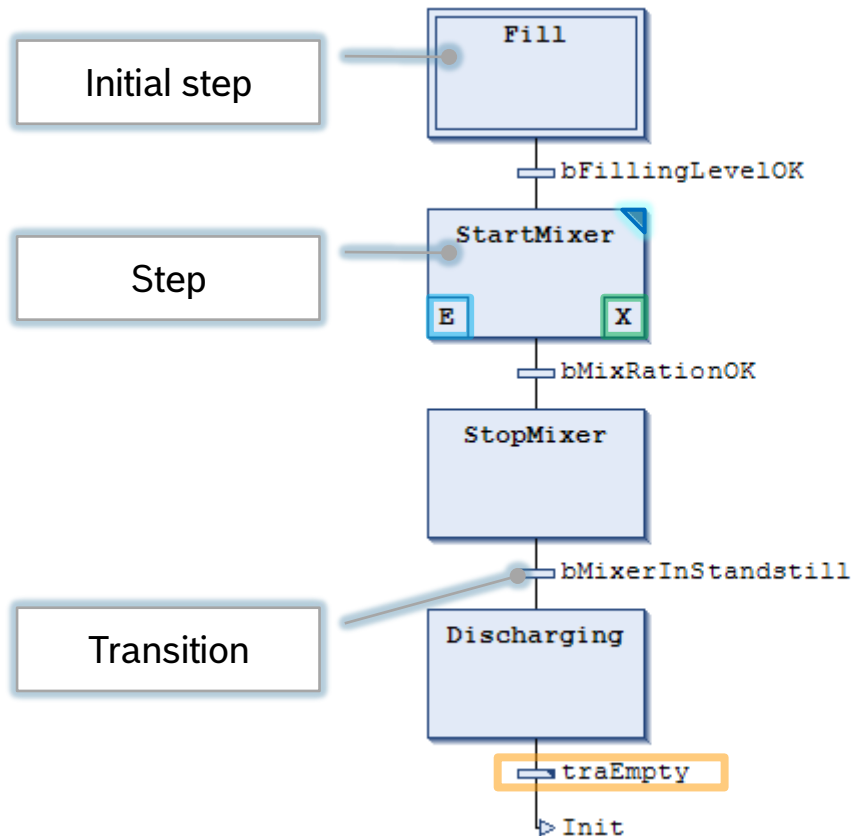
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SFC – Sequential Function Chart – Example



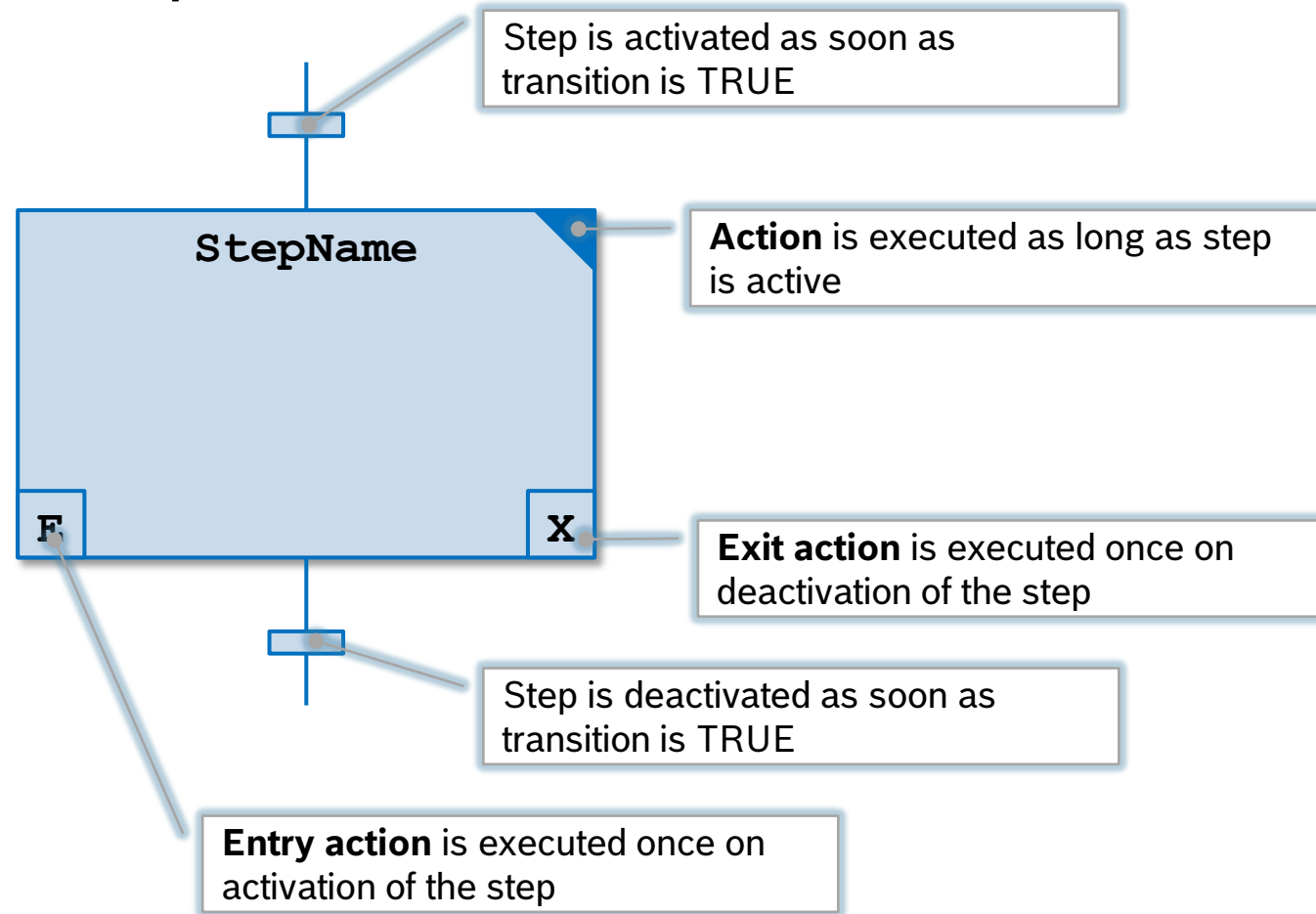
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SFC – Sequential Function Chart



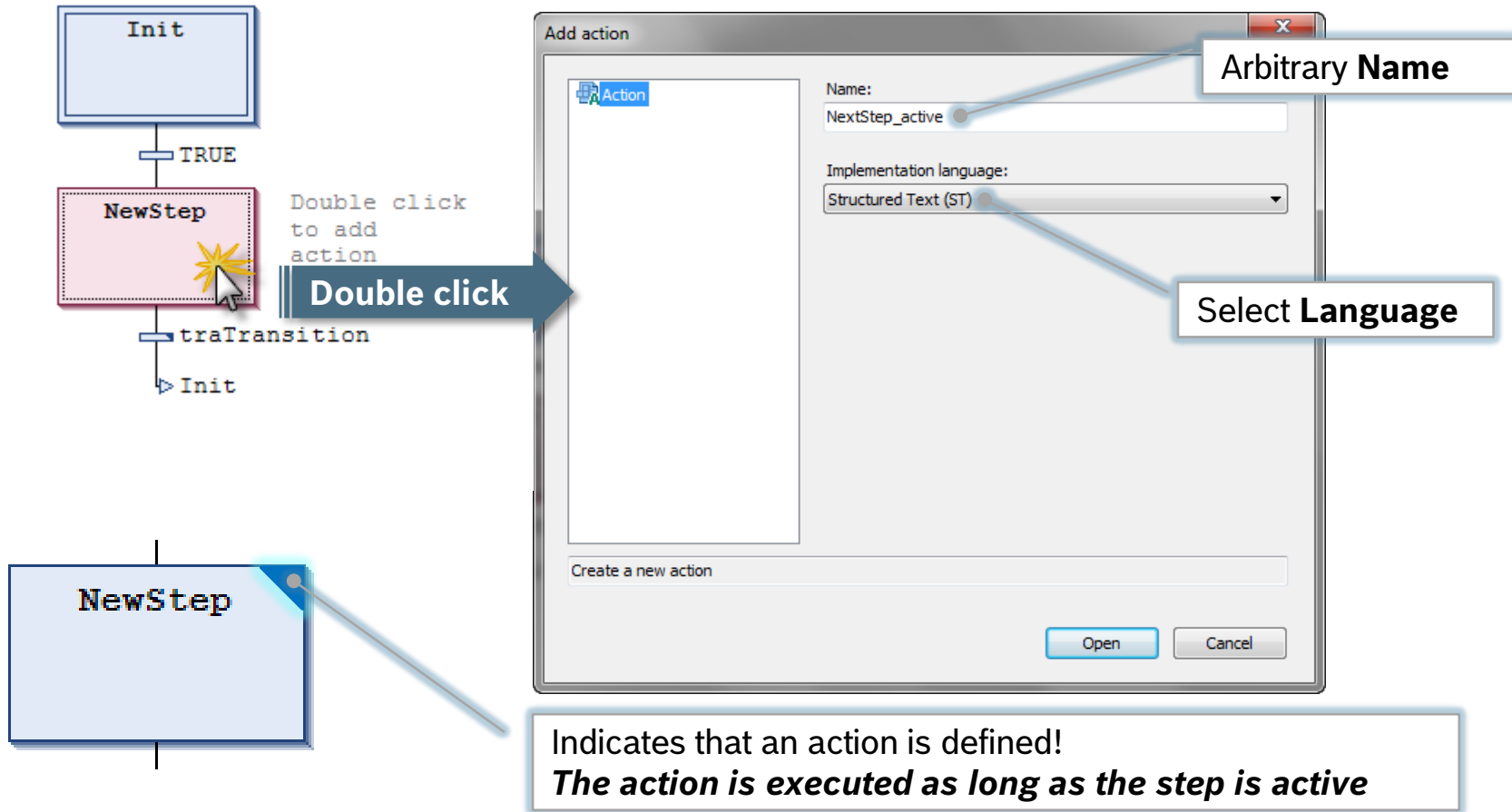
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SFC – Step



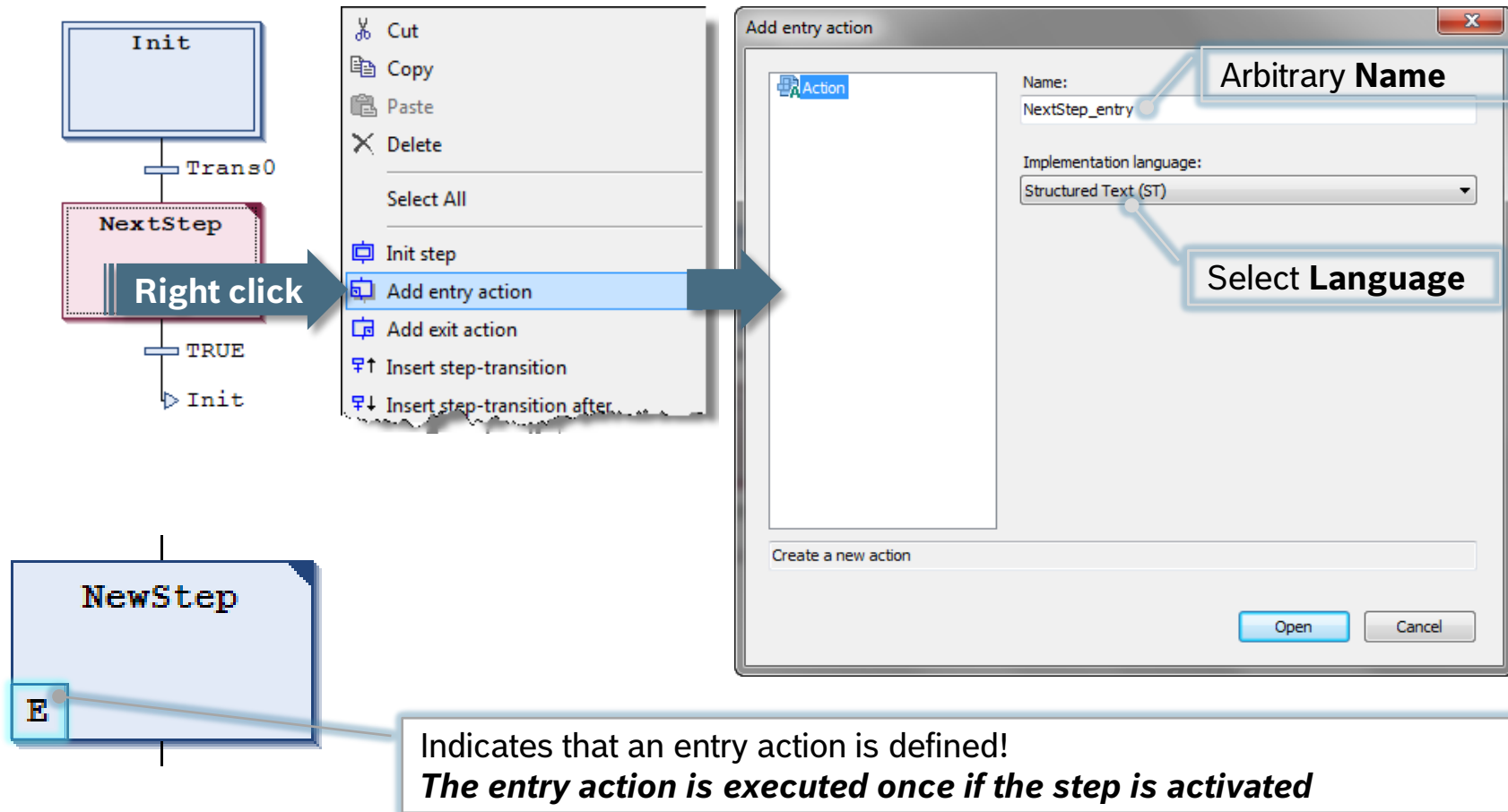
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SFC – How to add an action to a step



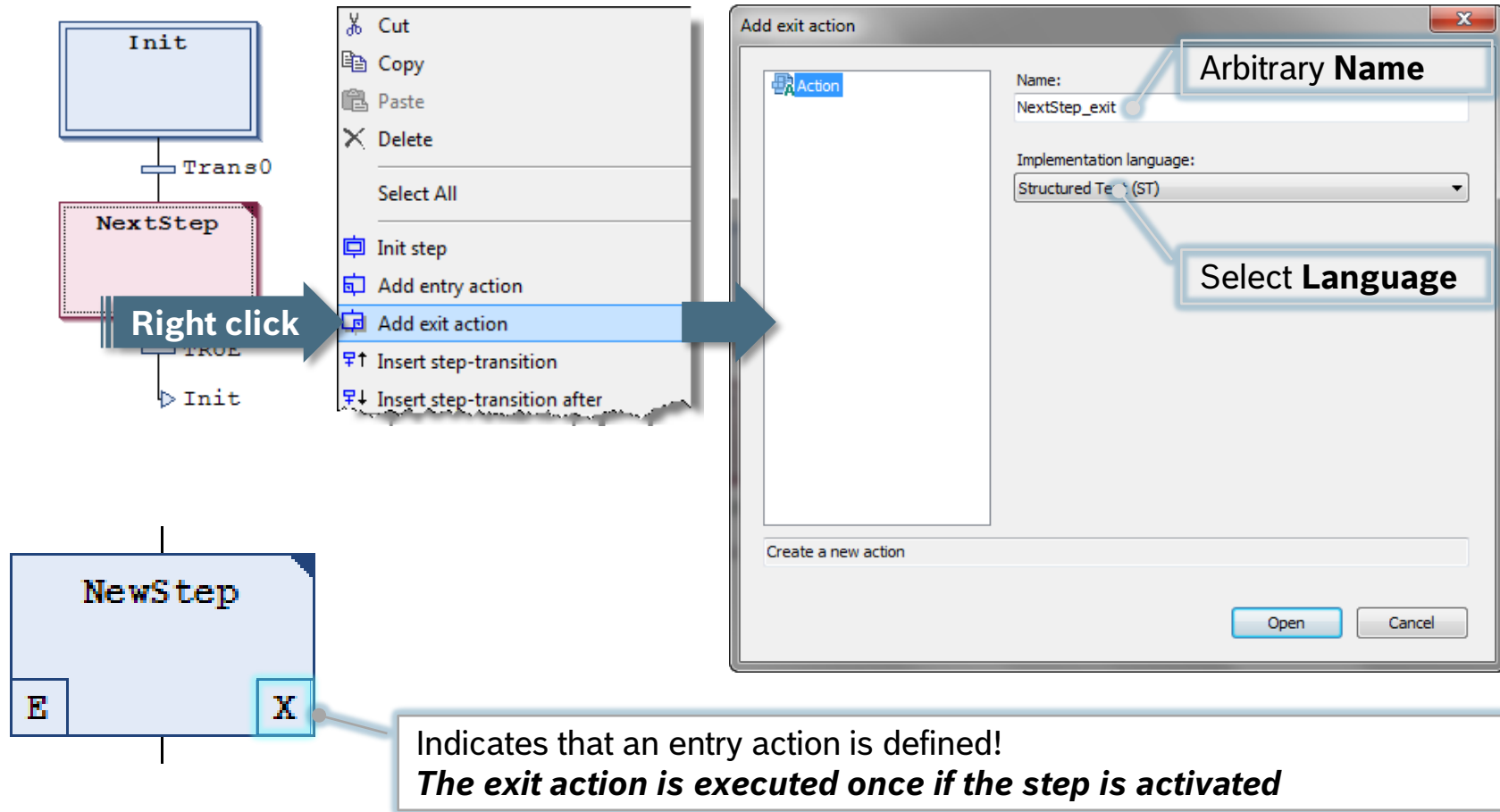
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SFC – How to add an entry action to a step



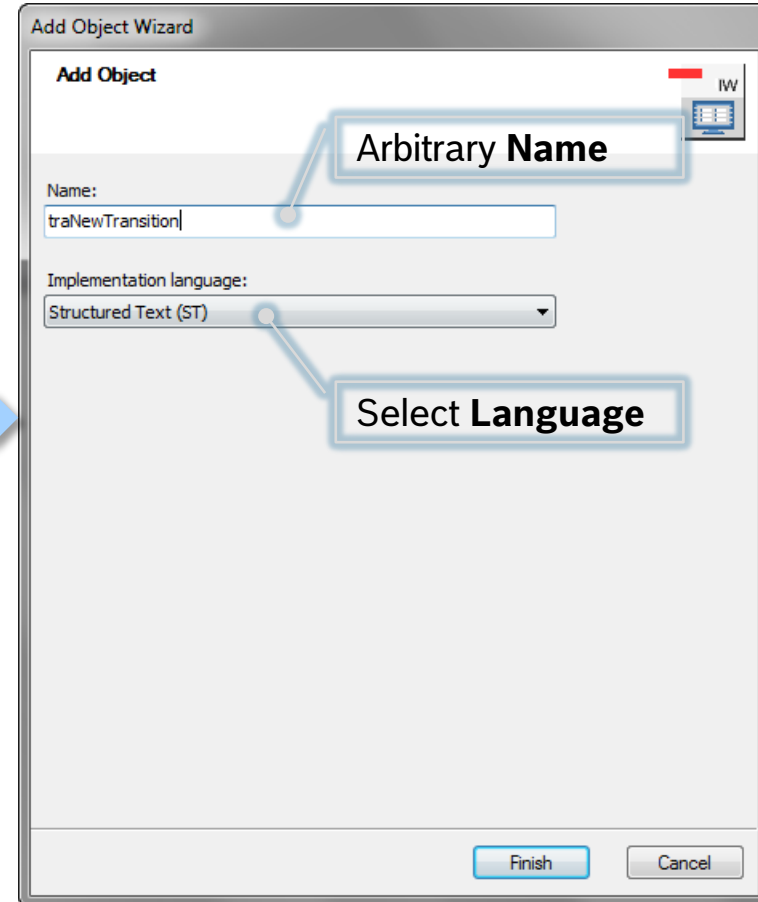
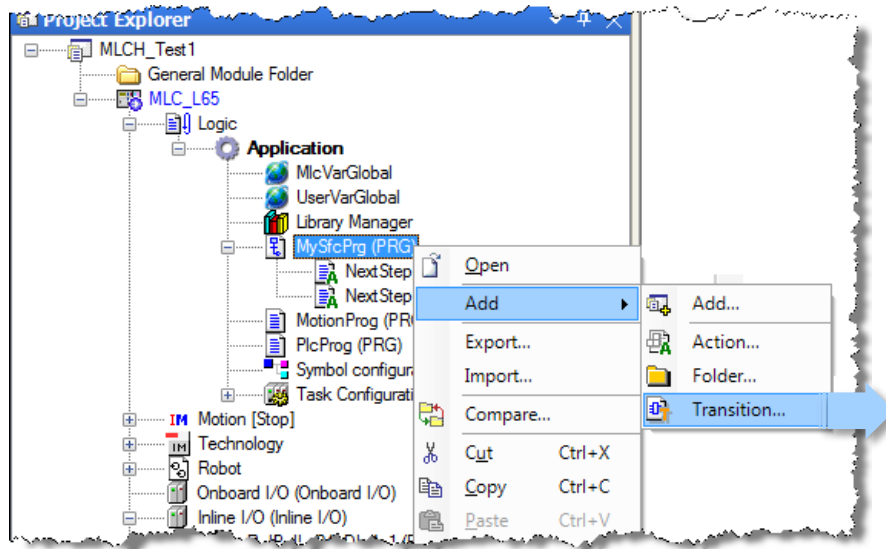
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SFC – How to add an exit action to a step



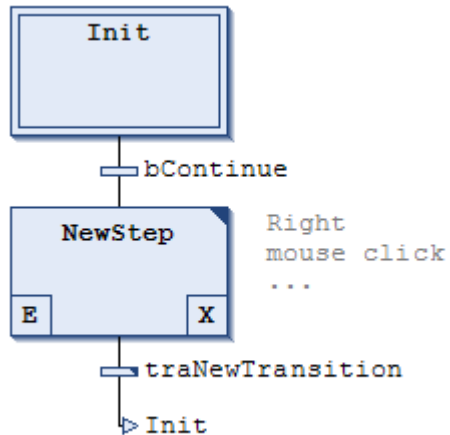
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SFC – Transition



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SFC – Transition

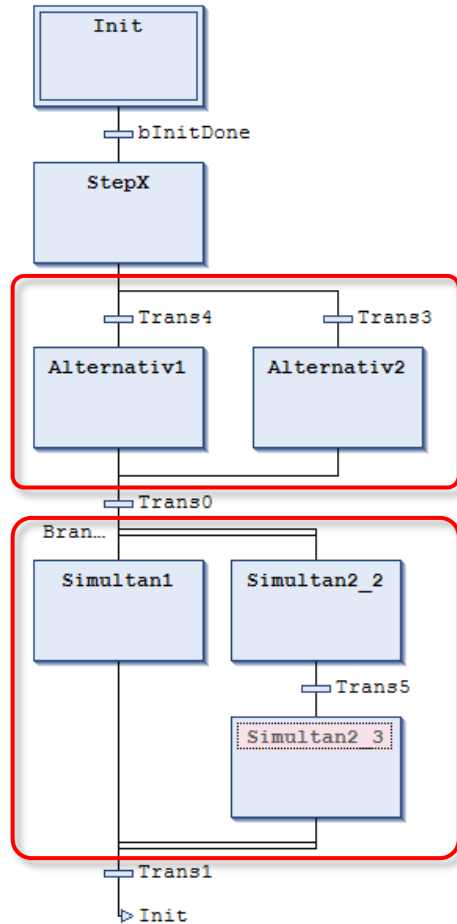


Direct statement of the transition condition
(Boolean expression)

Transition Name
→ Transition is an algorithm which returns a Boolean value

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SFC – Alternative and Parallel Branches



- After a step one branch or more branches can follow
- These branches can be alternative or parallel

Alternative branches

- If Trans4 changes to TRUE, StepX is deactivated and the step Alternativ1 is activated
- If Trans3 changes to TRUE, StepX is deactivated and the step Alternativ2 is activated

Simultaneous branches

- If Trans0 changes to TRUE, the step Alternativ1 or Alternativ2 are deactivated and the steps Simultan1 and Simultan2_2 are activated
- If Simultan1 and Simultan2_3 are active and Trans1 changes to TRUE, both steps are deactivated and the next step will be active